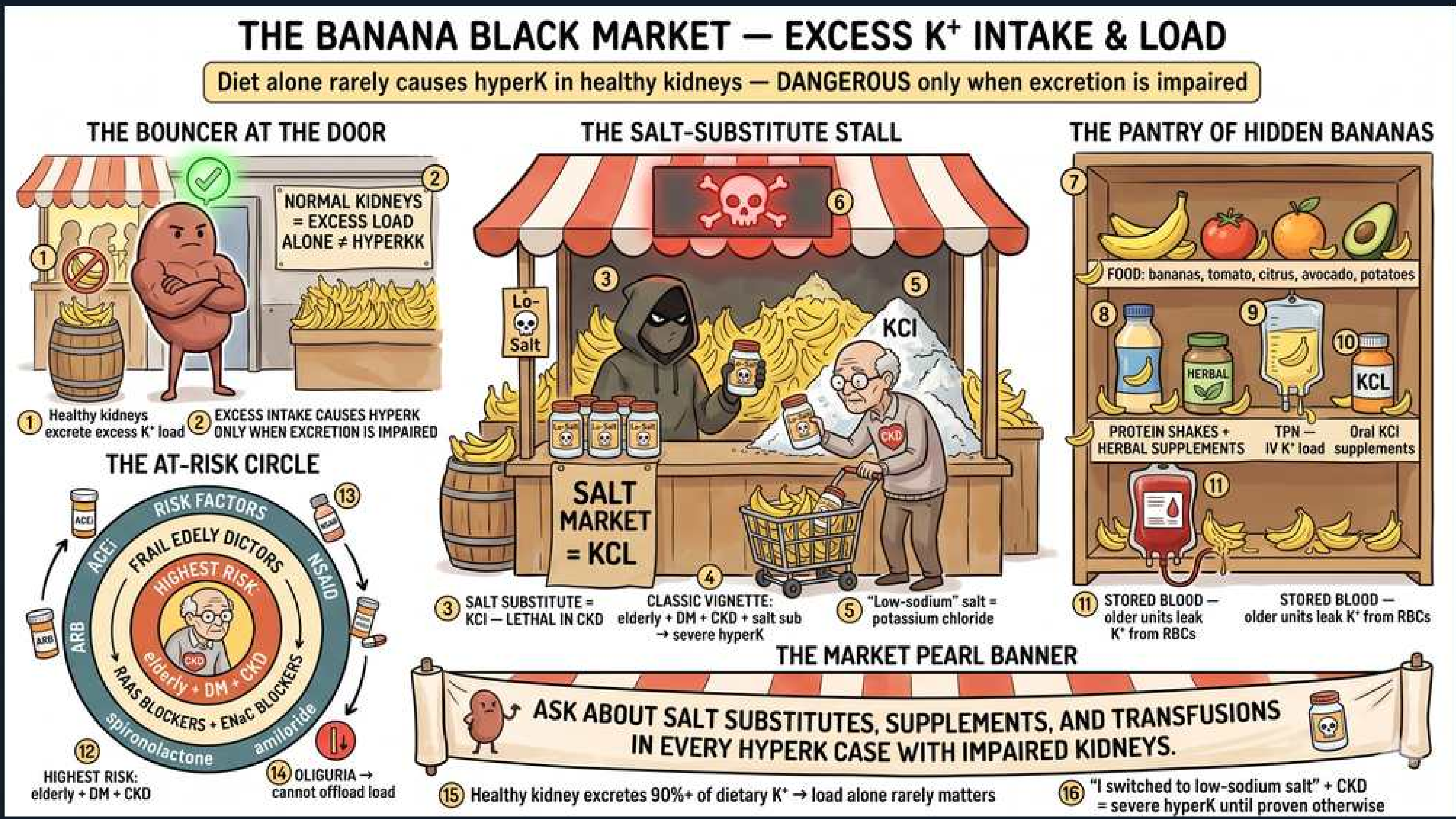
Hyperkalemia — excess intake & load



1. Healthy kidney excretes

WHAT YOU SEE

THE BOUNCER AT THE DOOR — a burly doorman waving a green check, turning away the potassium load — healthy kidneys excrete ~90% of dietary K+, so intake alone rarely overwhelms them.

WHAT IT ENCODES

Healthy kidneys have enormous capacity to excrete potassium — they offload ~90% of dietary K+ and can adapt to large chronic loads via aldosterone-driven distal secretion. Because of this reserve, increased intake alone almost never causes sustained hyperkalemia when renal function and aldosterone are intact. The body can also acutely buffer a load by shifting K+ into cells (insulin, beta-2 stimulation).

BOARD TRAP

WRONG ANSWER: attributing hyperkalemia in a patient with normal kidneys to a high-potassium diet. Discriminator: with normal GFR and aldosterone, dietary load rarely sustains hyperkalemia — look instead for pseudohyperkalemia, a transcellular shift, or an occult excretion defect (drugs, hypoaldosteronism) before blaming food.

2. Load + impaired excretion

WHAT YOU SEE

Beside the bouncer, the sign 'excess intake causes hyperK ONLY when excretion is impaired' — the two-hit rule: a hidden load PLUS a clearance defect.

WHAT IT ENCODES

Excess potassium load causes true hyperkalemia mainly when excretion is already impaired — CKD/AKI, hypoaldosteronism (Type 4 RTA), or K+-retaining drugs. The combination of a hidden load (salt substitute, supplement, transfusion) plus reduced clearance is the classic two-hit setup. Always look for BOTH an excessive input and a clearance defect.

BOARD TRAP

WRONG to treat the K+ load and ignore the excretion defect, or vice versa. Discriminator: the lethal vignette is always a two-hit picture (e.g., CKD + ACEi + salt substitute) — identify and remove the load AND address the clearance problem, not just one.

3. Salt substitute = KCl

WHAT YOU SEE

THE SALT-SUBSTITUTE STALL with a grinning grim-reaper vendor behind jars literally relabeled 'SALT = KCl' — the deadly seller pictures salt substitute as disguised potassium chloride.

WHAT IT ENCODES

Salt substitutes are potassium chloride (KCl) sold to replace sodium for hypertension/heart-failure patients — a classic hidden potassium source. A teaspoon can deliver 10-15 mEq of K+, and patients rarely volunteer it as a 'medication.' This is among the most commonly tested occult causes of hyperkalemia.

BOARD TRAP

WRONG to overlook the salt substitute because the patient denies taking potassium pills or eating bananas. Discriminator: salt substitutes/'lite salt' are KCl but are perceived as food, not medicine — you must specifically ask; the highest-yield victim is the elderly CKD/DM patient on an ACEi who switched salts.

4. Low-sodium salt

WHAT YOU SEE

A 'LOW-SODIUM SALT' container on the stall shelf marked KCl — the heart-healthy label hides a potassium load.

WHAT IT ENCODES

'Low-sodium,' 'lite,' or 'no-salt' seasonings are KCl-based by design — they substitute potassium for sodium. In CKD or on RAAS blockade they constitute a dangerous K+ load. The marketing as heart-healthy is exactly why high-risk cardiac/renal patients adopt them.

BOARD TRAP

WRONG to assume a 'low-sodium' product is K+-safe because it is marketed for hypertension and heart disease. Discriminator: low-sodium = high-potassium in these products; the very patients told to cut sodium (CHF, CKD on ACEi/ARB) are those least able to handle the KCl load.

5. Classic vignette

WHAT YOU SEE

The elderly customer at the stall buying salt substitute, tagged elderly + DM + CKD + ACEi — the board-classic stack of risk factors converging on one shopper.

WHAT IT ENCODES

The board-classic vignette: an elderly man with diabetes and CKD, on an ACEi or ARB, switches to a salt substitute and presents with severe hyperkalemia and ECG changes. Each factor — age, DM, CKD, RAAS blockade, KCl load — independently impairs K+ handling, and they stack. Recognizing the composite picture earns the diagnosis quickly.

BOARD TRAP

WRONG to chase a single dramatic cause (e.g., rhabdo, TLS) when the stem quietly stacks elderly + DM + CKD + ACEi + salt substitute. Discriminator: when multiple modest risk factors co-occur, the answer is the cumulative load-plus-impaired-excretion picture, not one exotic etiology.

6. Dietary K+ foods

WHAT YOU SEE

THE PANTRY OF HIDDEN BANANAS — shelves of bananas, tomatoes, citrus, avocado, potatoes — high-K+ foods that matter only once excretion is impaired.

WHAT IT ENCODES

High-potassium foods include bananas, tomatoes/tomato sauce, citrus and orange juice, avocado, potatoes, dried fruit, beans, and nuts. Dietary counseling matters in CKD/dialysis patients whose clearance is fixed. Still, in someone with normal kidneys these foods alone do not sustain hyperkalemia.

BOARD TRAP

WRONG to pick 'dietary potassium' as the cause of hyperkalemia in a patient with preserved renal function. Discriminator: food becomes the culprit only when excretion is impaired (CKD, hypoaldosteronism); with normal kidneys, look elsewhere first.

7. Supplements/herbal

WHAT YOU SEE

Protein-powder and herbal-supplement bottles in the pantry — hidden K+ sources patients don't count as 'medications.'

WHAT IT ENCODES

Protein powders, herbal products (e.g., noni juice, alfalfa, dandelion), and 'electrolyte' or coconut-water products can carry hidden potassium. Patients often do not consider supplements to be drugs and omit them from the history. In CKD this hidden load can tip a patient into dangerous hyperkalemia.

BOARD TRAP

WRONG to accept a 'no medications' history at face value in a hyperkalemic CKD patient. Discriminator: supplements and herbals are frequently underreported K+ sources — explicitly ask about powders, herbal teas, and 'natural' products separate from prescription meds.

8. TPN / oral KCl

WHAT YOU SEE

A hanging TPN bag and a bottle of oral KCl pills — iatrogenic potassium that overshoots when GFR drops.

WHAT IT ENCODES

Iatrogenic potassium is common: TPN/IV fluids with added K+, oral KCl supplements (often prescribed for diuretic-induced hypokalemia), and K+-containing penicillin. Inpatients on repletion protocols can overshoot, especially when renal function declines. Always reconcile ordered potassium against current GFR and serum K+.

BOARD TRAP

WRONG to continue standing KCl repletion or K+-containing IVF when GFR drops or oliguria develops. Discriminator: a repletion order appropriate yesterday becomes dangerous with new AKI — review iatrogenic K+ sources whenever renal function changes.

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9. Stored blood

WHAT YOU SEE

A bag of STORED BLOOD with K+ leaking out of aging RBCs — older units carry a higher extracellular potassium load (dangerous in massive transfusion).

WHAT IT ENCODES

Stored (packed) red cells leak potassium from RBCs over their shelf life, so older units carry a higher extracellular K+ load. Massive or rapid transfusion — trauma, exchange transfusion, irradiated units — can deliver enough K+ to cause acute hyperkalemia. Risk is amplified in renal failure and in neonates.

BOARD TRAP

WRONG to overlook transfusion as a hyperkalemia source after major resuscitation, especially in renal failure. Discriminator: massive/rapid transfusion of older or irradiated blood is a classic cause; for high-risk patients use fresher or washed units and monitor K+ during transfusion.

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10. At-risk circle

WHAT YOU SEE

THE AT-RISK CIRCLE wheel with 'HIGHEST RISK' at the bullseye (elderly + DM + CKD) — the center of the target marks the lowest-reserve patient.

WHAT IT ENCODES

Highest combined risk sits at the center of the wheel: elderly + diabetes + CKD, which together blunt the renin-aldosterone axis and distal K+ secretion. Diabetes contributes hyporeninemic hypoaldosteronism (Type 4 RTA), and aging plus CKD lower nephron mass and GFR. Layering K+-retaining drugs on this substrate is what precipitates crises.

BOARD TRAP

WRONG to view each risk factor as individually low-risk and dismiss the patient. Discriminator: risk is multiplicative — the elderly diabetic with CKD has the least reserve, so even a modest load or a single new drug (ACEi, spironolactone, NSAID) can be lethal.

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11. ACEi/ARB risk

WHAT YOU SEE

The 'ACE blockers / ARB blockers' wedge of the risk wheel — RAAS blockade dropping aldosterone, a top outpatient cause.

WHAT IT ENCODES

ACE inhibitors and ARBs reduce angiotensin II-driven aldosterone, decreasing distal K+ secretion and promoting retention. Risk rises in CKD, diabetes, volume depletion, and when combined with other RAAS agents or NSAIDs. They are among the most common drug causes of outpatient hyperkalemia.

BOARD TRAP

WRONG to combine an ACEi with an ARB or with spironolactone for 'extra' renoprotection/BP control without watching K+. Discriminator: dual RAAS blockade markedly increases hyperkalemia (and AKI) risk with little added benefit — check K+ and creatinine 1-2 weeks after starting or up-titrating.

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12. K-sparing diuretic

WHAT YOU SEE

The K-SPARING DIURETIC wedge of the wheel (spironolactone/eplerenone + amiloride/triamterene) — drugs that retain K+ at the distal nephron.

WHAT IT ENCODES

Potassium-sparing diuretics impair distal K+ excretion: spironolactone/eplerenone block the mineralocorticoid receptor, while amiloride/triamterene block ENaC. By cutting distal Na+ reabsorption and aldosterone effect, they reduce the driving force for K+ secretion. On a background of CKD or RAAS blockade, they readily cause hyperkalemia.

BOARD TRAP

WRONG to add spironolactone to a heart-failure patient already on an ACEi/ARB without monitoring or to use it in advanced CKD freely. Discriminator: MRAs (spironolactone/eplerenone) and ENaC blockers (amiloride/triamterene) both retain K+ — avoid stacking with RAAS agents in CKD and monitor closely.

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13. Oliguria

WHAT YOU SEE

The OLIGURIA segment — a kidney that 'cannot offload' with little urine flow — no flow means no way to sweep secreted K+ away.

WHAT IT ENCODES

Oliguria removes the urine flow needed to sweep secreted K+ away and signals failing excretory capacity. Without adequate distal flow and Na+ delivery, the kidney cannot maintain the gradient for K+ secretion. In an oliguric patient, ANY potassium load — dietary, iatrogenic, transfused, or endogenous — becomes hazardous.

BOARD TRAP

WRONG to continue routine K+ intake/repletion in an oliguric patient assuming the kidney will compensate. Discriminator: oliguria means clearance is gone — definitive removal (often dialysis) may be required, and ongoing K+ inputs must be stopped.

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14. Endogenous cell lysis

WHAT YOU SEE

The internal-load banner up top: TLS, rhabdomyolysis, hemolysis spilling intracellular K+ into the blood — cell breakdown as a massive endogenous potassium dump.

WHAT IT ENCODES

A massive 'load' can be endogenous: cell breakdown dumps intracellular K+ into the blood. Tumor lysis syndrome (post-chemo, with hyperkalemia + hyperphosphatemia + hyperuricemia + hypocalcemia), rhabdomyolysis (crush injury, with high CK and myoglobinuria), and brisk hemolysis all release large K+ quantities. These are dangerous because the load is huge and continuous until the process is controlled.

BOARD TRAP

WRONG to limit the hyperkalemia differential to intake and drugs and miss internal release. Discriminator: TLS (look for high phosphate/uric acid, low calcium after chemo) and rhabdomyolysis (high CK, dark urine after crush/immobilization) are major endogenous loads — and renal injury from these compounds the problem by impairing excretion.

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15. Ask about sources

WHAT YOU SEE

THE MARKET PEARL BANNER stretched across the bottom: 'ask about salt substitutes, supplements, and transfusions in every hyperK case with impaired kidneys' — the take-home to hunt hidden inputs plus a clearance defect.

WHAT IT ENCODES

In every hyperkalemia case, take a deliberate K+-source history: salt substitutes/'lite' salt, supplements and herbals, oral or IV KCl, recent transfusions, and high-K+ foods — plus K+-retaining drugs (ACEi/ARB, MRA, ENaC blockers, NSAIDs, trimethoprim, heparin). Then pair the input search with an excretion assessment (GFR, urine output, aldosterone status). This input-plus-clearance framework catches the two-hit cases boards favor.

BOARD TRAP

WRONG to stop the history at prescription medications and standard diet. Discriminator: the tested causes are precisely the easily missed ones — salt substitutes, supplements, and stored-blood transfusions — so ask about all hidden sources in addition to checking renal/aldosterone clearance.